# RentersEnergy

## A user-friendly electricity management tool for Queensland renters

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# Project Overview

RentersEnergy is a responsive website designed to help renters understand and manage their electricity costs. Many renters struggle with unclear bills, reactive management, and feeling excluded from homeowner-focused programs. Our solution provides:

- Clear visual breakdowns of electricity usage
- Proactive alerts for abnormal consumption
- Renter-specific energy saving tips
- Easy appliance configuration
- Government support program information

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# Team OL1 – 2008ICT Design Thinking in IT

| Name | Role |
|---|---|
| Yuehui Chen | Team Lead, Home page, CSS |
| Thomas Benfer | Alerts page |
| Jack Carrall | Usage page, Pitch script |
| Kane Bebb | Settings page |
| David Acworth | Help page |
| Ashley Burgoyne | Pitch video recording |

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# Tech Stack

- **HTML5** – Semantic structure
- **CSS3** – Flexbox, Grid, Variables, Media Queries
- **No JavaScript** – Static prototype (hard-coded mock data)
- **No backend** – All data is mock for demonstration

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# Features

## 5 Screens

| Screen | Description |
|---|---|
| **Home** | Dashboard with alerts, usage chart, government support card |
| **Alerts** | Master-detail layout, bulk actions, notification settings |
| **Electricity Usage** | Appliance and room breakdown, period selector |
| **Settings** | Profile, property, and appliance configuration |
| **Help** | FAQ, contact options, live chat mock |

## Design Highlights

- **Mobile-first responsive design** – Bottom navigation on mobile, top nav on desktop
- **Colourblind-friendly palette** – Blue and amber tones (no red-green)
- **Accessibility** – WCAG contrast, keyboard focus outlines, alt text on images
- **CSS animations** – Pulsing NEW badge, button hover effects

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# How to Run

1. Download the ZIP file

2. Extract all files to a folder
3. Open `index.html` in a web browser
4. No server or installation required

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# File Structure

```
rentersenergy/
├── index.html (Home page)
├── alerts.html (Alerts page)
├── usage.html (Electricity Usage page)
├── settings.html (Settings page)
├── help.html (Help page)
├── css/
│   ├── reset.css (CSS reset)
│   └── style.css (All styles)
├── images/ (Logo, charts, icons)
├── .gitignore
└── README.md
```

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# Colour Palette

| Colour | Hex | Usage |
|---|---|---|
| Primary (Teal) | #4A9B7A | Headers, main nav, save buttons |
| Secondary (Blue) | #0474cc | Edit, Add New, dropdown buttons |
| Warning (Amber) | #D97706 | Warnings, high usage alerts |
| Delete (Burgundy) | #97233F | Delete account, cancel |
| Highlight (Yellow) | #FFF3B0 | Unread alert background |

## Chart Palette (Electricity-themed)

| Usage Level | Colour | Hex |
|---|---|---|
| Low | Power Green | #32CD32 |
| Medium | Electric Blue | #00BFFF |
| High | Surge Orange | #FF8C00 |
| Critical | Voltage Red | #FF4444 |

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# Security Considerations

## Current Prototype

- Limited free-text input (minimal attack surface)
- Confirmation dialogs for destructive actions
- Only necessary data collection (no sensitive personal data)

## Production Future

- HTTPS with TLS 1.3
- Backend authentication (JWT in HTTP-only cookies)
- Encrypted database storage
- Input sanitisation and validation
- Rate limiting on login attempts

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# Testing

The website was tested using:

- **Cognitive walkthrough** with other student teams
- **Mobile-first testing** (Chrome DevTools device emulation)
- **Cross-browser testing** (Chrome, Firefox, Edge)
- **Accessibility checks** (keyboard navigation, contrast)

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## Credits

- Icons: Emoji and Unicode symbols
- Colour palette: Custom, based on WCAG accessibility guidelines
- Font: System default (system-ui) for performance

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## Submission

- **Due date:** Friday Week 12
- **Course:** 2008ICT Design Thinking in IT
- **Institution:** Griffith University

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